

Mathematics Standard 11–12 (2024)

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Mathematics Standard 11–12 (2024)

Implementation from 2026

The new Mathematics Standard 11–12 Syllabus (2024) is to be implemented from 2026 and will replace the [Mathematics Standard Stage 6 Syllabus \(2017\)](#).

2026, Term 1

- Start teaching the new syllabus for Year 11
- Start implementing the new Year 11 school-based assessment requirements
- Continue to teach the [Mathematics Standard Stage 6 Syllabus \(2017\)](#) for Year 12

2026, Term 4

- Start teaching the new syllabus for Year 12
- Start implementing the new Year 12 school-based assessment requirements

2027

- First HSC examination for the new syllabus

Overview

Syllabus overview

The *Mathematics Standard 11–12 Syllabus* aligns with the *Mathematics Life Skills 11–12 Syllabus* to provide opportunities for integrated delivery.

Through [collaborative curriculum planning](#), it may be decided that [Life Skills outcomes and content](#) are the most appropriate option for some students with intellectual disability.

Organisation of Mathematics Standard 11–12

The organisation of outcomes and content for Mathematics Standard 11–12 highlights the important role Working mathematically plays across all areas of mathematics and reflects the strengthened connections between concepts. Working mathematically has been embedded in the outcomes and content of the syllabus.

Mathematics Standard 11–12 outcomes and their related content are organised into 5 areas of study:

- Algebra
- Financial mathematics
- Measurement
- Networks
- Statistics.

Figure 1 shows the organisation of Mathematics Standard 11–12. All Year 11 Mathematics Standard focus areas are to be completed first, then focus areas from Year 12 Mathematics Standard 1 OR Year 12 Mathematics Standard 2 are to be completed.

+ Mathematics Standard 11–12			
Working mathematically through communicating reasoning, understanding and fluency, and problem solving			
	Year 11 Mathematics Standard	Year 12 Mathematics Standard 1	or Year 12 Mathematics Standard 2
Algebra	Formulas and equations Linear relationships	Algebraic relationships	Algebraic relationships
Financial mathematics	Earning money Managing money	Investment Depreciation and loans	Investment and loans Annuities
Measurement	Applications of measurement Time and location	Right-angled triangles Ratios and rates	Trigonometry Ratios and rates
Networks	Networks, paths and trees		Network flow Critical path analysis
Statistics	Data analysis	Bivariate data analysis Relative frequency and probability	Bivariate data analysis Relative frequency and probability The normal distribution

Figure 1: The organisation of Mathematics Standard 11–12

Image long description: Five rows represent the areas of study for Mathematics Standard 11–12. Each row is divided into thirds, showing the focus areas for Year 11, Year 12 (Mathematics Standard 1) or Year 12 (Mathematics Standard 2). The Algebra area of study includes: Year 11 focus areas Formulas and equations and Linear relationships; the Year 12 (Standard 1) focus area Algebraic relationships; and the Year 12 (Standard 2) focus area Algebraic relationships. The Financial mathematics area of study includes: the Year 11 focus areas Earning money, and Managing money; the Year 12 (Standard 1) focus areas Investment, and Depreciation and loans; and the Year 12 (Standard 2) focus areas Investment and loans, and Annuities. The Measurement area of study includes: the Year 11 focus areas Applications of measurement, and Time and location; the Year 12 (Standard 1) focus areas Right-angled triangles, and Ratios and rates; and the Year 12 (Standard 2) focus areas Trigonometry, and Ratios and rates. The Networks area of study includes: the Year 11 focus area Networks, paths and trees; and the Year 12 (Standard 2) focus areas Network flow, and Critical path analysis. The Statistics area of study includes: the Year 11 focus area Data analysis; the Year 12 (Standard 1) focus areas Bivariate data analysis, and Relative frequency and probability; and the Year 12 (Standard 2) focus areas Bivariate data analysis, Relative frequency and probability, and The normal distribution. All content is surrounded by a box labelled with the phrase, ‘Working mathematically through communicating reasoning, understanding and fluency, and problem solving’.

Protocols for collaborating with Aboriginal and Torres Strait Islander Communities

NESA is committed to working in partnership with Aboriginal Communities and supporting teachers, schools and schooling sectors to improve educational outcomes for young people.

It is important to respect appropriate ways of interacting with Aboriginal Communities and Cultural material when teachers plan, program and implement learning experiences that focus on Aboriginal and Torres Strait Islander Priorities.

Indigenous Cultural and Intellectual Property (ICIP) protocols need to be followed. Aboriginal and Torres Strait Islander Peoples’ ICIP protocols include Cultural Knowledges, Cultural Expression and Cultural Property and documentation of Aboriginal and Torres Strait Islander Peoples’

Identities and lived experiences. It is important to recognise the diversity and complexity of different Cultural groups in NSW, as protocols may differ between local Aboriginal Communities.

Teachers should work in partnership with Elders, parents, Community members, Cultural Knowledge Holders, or a local, regional or state Aboriginal Education Consultative Group. It is important to respect Elders and the roles of men and women. Local Aboriginal Peoples should be invited to share their Cultural Knowledges with students and staff when engaging with Aboriginal histories and Cultural Practices.

Course description

Mathematics Standard 11–12 focuses on enabling students to use mathematics to make informed decisions in their daily lives. Students develop understanding and competence through real-world applications of mathematics.

Mathematics Standard 1 provides opportunities for students to build confidence and make mathematics meaningful. Students develop their mathematical knowledge and understanding through applying and modelling to prepare for post-school employment or further training.

Mathematics Standard 2 provides a pathway for students to extend their mathematical thinking by examining more complex content, and through applications and modelling.

What students learn

Through the study of Mathematics Standard 1, students:

- develop their knowledge, understanding and skills in Working mathematically and in communicating concisely and systematically
- consider various applications of mathematics in a broad range of contemporary contexts through mathematical modelling and use these models to solve problems related to their present and future needs
- gain an appropriate mathematical background for post-school employment or further training.

Through the study of Mathematics Standard 2, students:

- develop their knowledge, understanding and skills in Working mathematically and in communicating concisely and systematically
- consider various applications of mathematics in a broad range of contemporary contexts through mathematical modelling and use these models to solve problems related to their present and future needs
- develop an understanding of, and skills in, further aspects of mathematics for concurrent HSC studies
- gain an appropriate mathematical background for a wide range of educational and employment aspirations.

Course structure and requirements

Year 11 course structure

Area of study	Focus area
Algebra	Formulas and equations Linear relationships
Financial mathematics	Earning money Managing money
Measurement	Applications of measurement Time and location
Networks	Networks, paths and trees
Statistics	Data analysis

Year 12 Standard 1 course structure

Area of study	Focus area
Algebra	Algebraic relationships
Financial mathematics	Investment Depreciation and loans
Measurement	Right-angled triangles Ratios and rates
Statistics	Bivariate data analysis Relative frequency and probability

Year 12 Standard 2 course structure

Area of study	Focus area
Algebra	Algebraic relationships
Financial mathematics	Investment and loans Annuities
Measurement	Trigonometry Ratios and rates
Networks	Network flow Critical path analysis
Statistics	Bivariate data analysis

	Relative frequency and probability The normal distribution
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Safety and risk management

Schools are required to ensure they follow [safety and risk management](#) in delivering the *Mathematics Standard 11–12 Syllabus*.

Course enrolment details

Mathematics Standard Year 11

- Course number: 11236
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

Exclusions

- Mathematics Advanced (Year 11, 2 units): 11255
- Mathematics Extension 1 (Year 11, 1 unit): 11250
- Mathematics Life Skills (Year 11, 2 units): 17680

Prerequisites

- Nil

Corequisites

- Nil

Mathematics Standard 1 Year 12

- Course number: 15231
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

Exclusions

- Mathematics Standard 1 (examination) (Year 12, 2 units): 15232
- Mathematics Standard 2 (Year 12, 2 units): 15236
- Mathematics Advanced (Year 12, 2 units): 15255
- Mathematics Extension 1 (Year 12, 1 unit): 15250
- Mathematics Extension 2 (Year 12, 1 unit): 15260
- Mathematics Life Skills (Year 12, 2 units): 17680

Prerequisites

- Mathematics Standard (Year 11, 2 units): 11236

Corequisites

- Nil

Mathematics Standard 2 Year 12

- Course number: 15236
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

Exclusions

- Mathematics Standard 1 (Year 12, 2 units): 15231
- Mathematics Standard 1 (examination) (Year 12, 2 units): 15232
- Mathematics Advanced (Year 12, 2 units): 15255
- Mathematics Extension 1 (Year 12, 1 unit): 15250
- Mathematics Extension 2 (Year 12, 1 unit): 15260
- Mathematics Life Skills (Year 12 , 2 units): 17680

Prerequisites

- Mathematics Standard (Year 11, 2 units): 11236

Corequisites

- Nil

HSC information

Information about [curriculum requirements](#) for the HSC are available on [Assessment Certification Examination \(ACE\)](#).

Rationale

Mathematics in the Stage 6 Curriculum

Mathematical ideas have evolved and continue to develop across cultures and have been practised in Australia by Aboriginal and Torres Strait Islander Peoples for thousands of years.

The utility of mathematics extends from counting and measuring to explaining real-world phenomena and its inherent beauty. It has evolved in sophisticated ways to become the language used to describe and understand much of the modern world.

The creative application of mathematical concepts is integral to scientific and technological advancement. The symbolic nature of mathematics provides a precise means of communication.

The Mathematics Stage 6 syllabuses are designed to offer students opportunities to think mathematically. Mathematical thinking is supported through questioning, communicating, reasoning, reflecting and the use of appropriate technology, and is encouraged through providing students with opportunities to generalise, challenge, find connections and to think critically and creatively. Using mathematical thinking, students apply their knowledge and skills to deepen their understanding of the world.

Mathematics Standard

The Mathematics Standard courses are focused on enabling students to use mathematics effectively, efficiently and critically to make informed decisions in their daily lives. They provide students with opportunities to develop an understanding of, and competence in, aspects of mathematics through real-world applications. Mathematics Standard Year 11 provides a pathway for students who progress through the Core outcomes of the *Mathematics 7–10 Syllabus* (2022). This course is designed for students who want to extend their mathematical skills beyond Stage 5, gain further knowledge of mathematical concepts and apply these skills and knowledge in practical contexts.

In Year 12, Mathematics Standard 1 provides an opportunity for students to continue to master aspects of the Year 11 outcomes and develop their mathematical knowledge and understanding through applications and modelling. This supports students to see the significance of mathematics and have the opportunity to prepare for post-school employment or further training.

Mathematics Standard 2 provides a pathway for students to extend their mathematical thinking through examining more complex content, and through applications and modelling. The Mathematics Standard 2 course offers students the opportunity to prepare for a wide range of educational and employment aspirations.

Aim

Mathematics Standard in Stage 6 enables students to develop their knowledge and understanding from Stage 5 of how to work mathematically, make connections within mathematics, use mathematical models, relate mathematical concepts to their world, and improve their application of mathematical language to communicate in a concise and systematic manner.

Table of outcomes

MAO-WM-01 Working mathematically

develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

This outcome is aligned to all content in each Stage.

Year 11	Year 12 – Standard 1	Year 12 – Standard 2
MST-11-01 selects and applies algebraic techniques to solve problems involving equations and formulas	MST-12-S1-01 represents and interprets the relationships between quantities in algebraic and graphical forms	MST-12-S2-01 represents the relationships between quantities algebraically and graphically to solve problems and make predictions in practical contexts
MST-11-02 models and interprets linear relationships to solve problems and make predictions in practical contexts	MST-12-S1-02 solves problems involving simple interest and compound interest to make decisions about financial situations	MST-12-S2-02 models financial situations and solves problems involving interest, depreciation and borrowing money
MST-11-03 solves financial problems involving earning money and taxation	MST-12-S1-03 solves problems involving interest and depreciation to make decisions about loans and credit cards	MST-12-S2-03 solves financial problems involving annuities
MST-11-04 solves financial problems involving budgeting and purchasing	MST-12-S1-04 applies Pythagoras' theorem and trigonometry to solve problems involving right-angled triangles	MST-12-S2-04 applies trigonometry to solve problems involving right-angled and non-right-angled triangles
MST-11-05 solves problems involving measurement in practical contexts	MST-12-S1-05 interprets ratios and rates to solve problems in practical contexts	MST-12-S2-05 solves practical problems involving ratios and rates
MST-11-06 solves problems involving time and location in practical contexts	MST-12-S1-06 displays and interprets bivariate datasets to solve problems and make predictions	MST-12-S2-06 applies network flow concepts to solve problems and model decision-making in practical contexts
MST-11-07 applies network techniques to solve network problems	MST-12-S1-07 models and solves problems involving the probabilities of simple and compound events	MST-12-S2-07 applies critical path analysis to solve problems and model decision-making in practical contexts
MST-11-08 displays and analyses datasets using summary		MST-12-S2-08 analyses bivariate datasets using statistical processes

Year 11	Year 12 – Standard 1	Year 12 – Standard 2
statistics and graphical representations		
		MST-12-S2-09 models and solves problems involving the probabilities of multistage events
		MST-12-S2-10 analyses normally distributed datasets using statistical processes

Outcomes and content for Year 11

Formulas and equations

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- selects and applies algebraic techniques to solve problems involving equations and formulas **MST-11-01**

Related Life Skills outcomes: MA-LS-01, MA-LS-02, MA-LS-03, MA-LS-04

Content

- Substitute numbers into linear and non-linear [algebraic expressions](#), [equations](#) and formulas
- Evaluate the subject of a formula, given the value of other [pronumerals](#) in the formula
- Calculate [distance](#), [speed](#) and time (with change of units of measurement as required) and stopping distances of vehicles using $speed = \frac{distance}{time}$ and $stopping\ distance = reaction\ distance + braking\ distance$
- Apply the formulas $BAC_{male} = \frac{10N - 7.5H}{6.8M}$ and $BAC_{female} = \frac{10N - 7.5H}{5.5M}$ to calculate and interpret [blood alcohol content \(BAC\)](#) based on drink consumption and body weight, where N is the number of standard drinks consumed, H is the number of hours of drinking and M is the person's weight in kilograms
- Apply the formula $Time = \frac{BAC}{0.015}$ to determine the number of hours required for a person to stop consuming alcohol in order to reach zero BAC
- Identify and explain the limitations of methods used to estimate BAC
- Calculate the required medication dosages for children and adults given age or weight, using Fried's, Young's and Clark's formula as appropriate
- Apply given formulas to solve problems in a variety of contexts
- Change the subject of linear and non-linear formulas limited to quadratics of the form $y = ax^2 + c$ to solve problems and make decisions about solutions in the context of the problem
- Represent a word problem as a [linear equation](#), solve the equation and interpret the solution in the context of the problem
- Use a spreadsheet to perform calculations involving formulas

Linear relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- models and interprets linear relationships to solve problems and make predictions in practical contexts **MST-11-02**

Related Life Skills outcomes: MA-LS-01, MA-LS-02, MA-LS-03, MA-LS-04

Content

Linear modelling

- Determine the y -intercept and [gradient](#) of a straight line given in graphical form
- Determine the [equation](#) of a straight line of the form $y = mx + c$ where m is the gradient and c is the y -intercept
- Determine the y -intercept and gradient from the equation of a straight line
- Construct a straight-line [graph](#), with and without using graphing applications
- Model [linear relationships](#) and interpret the features of those relationships in a variety of contexts, noting that the y -intercept is the vertical [intercept](#) and the gradient is the [rate of change](#)
- [Interpolate](#) and [extrapolate](#) from a [linear model](#) in a practical context
- Identify and describe the limitations of a linear model in a practical context
- Use a spreadsheet to model a linear relationship in a variety of contexts

Direct variation

- Recognise that a [direct variation](#) relationship of the form $y = kx$ produces a straight-line graph
- Explain that a direct variation relationship of the form $y = kx$ is a graph that passes through the origin with gradient k being the [constant of variation](#)
- Identify and represent direct variation of the form $y = kx$ from descriptions of situations in which one quantity varies directly with another quantity
- Evaluate k in equations of the form $y = kx$, given one pair of values for the [variables](#), and use the resulting formula to find other values of the variables to solve problems in a variety of contexts
- Graph and analyse equations of the form $y = kx$ to solve problems in a variety of contexts

Earning money

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving earning money and taxation **MST-11-03**

Related Life Skills outcomes: MA-LS-05, MA-LS-06

Content

Ways of earning

- Solve problems involving salaries
- Solve problems involving [wages](#), [penalty rates](#) for [overtime](#), weekends and public holidays, given an hourly [rate](#) of pay
- Calculate an individual's earnings from [commission](#), sliding scale commission, [piecework](#), [royalties](#), [government payments](#) and as a sole trader
- Calculate weekly, fortnightly, monthly and yearly earnings assuming 1 year = 52 weeks
- Calculate leave loading by finding a percentage of eligible normal pay
- Use a spreadsheet to model an employee's income including overtime, [salary](#), bonuses, commission, piecework and royalties

Taxation

- Determine [taxable income](#) by examining allowable [deductions](#)
- Determine [tax](#) payable by interpreting and applying information in tax tables
- Calculate the weekly, fortnightly or monthly tax to be deducted from a worker's pay under the Australian [Pay-As-You-Go \(PAYG\) taxation](#) system using tax tables
- Calculate the [Medicare levy](#) from taxable income
- Calculate [net earnings](#) after deductions, taxation and the Medicare levy
- Determine by calculation if more tax is payable or if a refund is owed after completing a [tax return](#)
- Use a spreadsheet to model scenarios involving taxation

Managing money

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving budgeting and purchasing **MST-11-04**

Related Life Skills outcomes: MA-LS-05, MA-LS-06

Content

Purchasing goods

- Apply [percentage increase](#) and [decrease](#) to determine the [cost](#) of goods, and [profit](#) and [loss](#)
- Calculate the [goods and service tax \(GST\)](#) or [value-added tax \(VAT\)](#) when purchasing goods and services in Australia and internationally
- Solve problems involving GST or VAT
- Examine payment options involving [buy now, pay later](#) and analyse the costs associated with these schemes for purchasing goods
- Calculate the cost of buying on terms that involve paying an initial [deposit](#) and making regular [repayments](#)
- Compare the total cost of using different methods of purchasing goods and justify which method of purchasing is best suited to a given context
- Calculate the cost of purchasing a vehicle by considering the purchase price or [loan](#) repayments, registration, compulsory and non-compulsory third-party insurance, comprehensive insurance and [stamp duty](#) at current rates

Budgeting

- Examine the ongoing costs of a vehicle
- Use a spreadsheet to model the costs of purchasing, running and maintaining a vehicle
- Calculate household costs including electricity, water and gas usage from household bills
- Prepare a personal [budget](#) for a given income, considering [fixed](#) and [discretionary spending](#)
- Use a spreadsheet to model personal and household budgets
- Examine budgeting, saving, establishing an emergency fund with 3 to 6 months of living expenses, and monitoring spending as strategies designed to minimise financial problems

Applications of measurement

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving measurement in practical contexts **MST-11-05**

Related Life Skills outcomes: MA-LS-07, MA-LS-08, MA-LS-09, MA-LS-10

Content

Practicalities of measurement

- Identify and convert between the metric units of length: millimetres (mm), centimetres (cm), metres (m) and kilometres (km)
- Identify and convert between metric units of area, using $1 \text{ cm}^2 = 100 \text{ mm}^2$, $1 \text{ m}^2 = 10\,000 \text{ cm}^2$, $1 \text{ ha} = 10\,000 \text{ m}^2$ and $1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$
- Identify and convert between metric units of [volume](#), using $1 \text{ cm}^3 = 1000 \text{ mm}^3$, $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$, and $1 \text{ km}^3 = 1\,000\,000\,000 \text{ m}^3$
- Identify and convert between the metric units of [capacity](#): millilitres (mL), litres (L), kilolitres (kL) and megalitres (ML)
- Identify and convert between units of volume and capacity, using $1 \text{ cm}^3 = 1 \text{ mL}$, $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$ and $1000 \text{ kL} = 1 \text{ ML}$
- Identify and convert between the metric units of [mass](#): milligrams (mg), grams (g), kilograms (kg) and tonnes (t)
- Apply [scientific notation](#) to represent numbers involving standard prefixes: nano- (n) for 10^{-9} , micro- (μ) for 10^{-6} , milli- (m) for 10^{-3} , centi- (c) for 10^{-2} , kilo- (k) for 10^3 , mega- (M) for 10^6 , giga- (G) for 10^9 and tera- (T) for 10^{12} , with and without a required number of [significant figures](#)

Perimeter, area and volume

- Solve practical problems requiring the calculation of perimeters and areas of triangles, rectangles, parallelograms, trapeziums, [circles](#), [sectors](#) of circles and [composite shapes](#) in a variety of contexts
- Apply [Pythagoras' theorem](#) to solve problems involving right-angled triangles
- Calculate perimeters and areas of irregularly shaped blocks of land by dissection into triangles, rectangles and trapeziums
- Solve practical problems involving the [surface area](#) of prisms, [cylinders](#), spheres and composite solids in a variety of contexts
- Solve practical problems involving volume and capacity of prisms, cylinders, spheres, cones, pyramids and composite solids in a variety of contexts
- Apply the [Trapezoidal rule](#) $A \approx \frac{h}{2}(d_f + d_l)$, where A is the area, d_f and d_l are the lengths of the [parallel](#) sides of the trapezium, d_f is the first length and d_l is the last length, and h is the [perpendicular distance](#) between them, to solve a variety of practical problems involving area, volume and capacity for up to four applications

Time and location

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving time and location in practical contexts **MST-11-06**

Related Life Skills outcomes: MA-LS-07, MA-LS-08, MA-LS-09, MA-LS-10

Content

Positions on the Earth's surface

- Identify a location on the Earth's surface using [latitude](#), [longitude](#) and [position coordinates](#), with and without using digital tools
- Apply $15^\circ = 1$ hour time difference to calculate the time differences between locations on Earth given the relevant longitudes

Time and time differences

- Convert between seconds (s), minutes (min) and hours (h)
- Represent and convert between 12-hour and 24-hour time formats
- Solve a variety of practical problems by calculating elapsed time
- Solve problems involving [time zones](#) in Australia, making necessary allowances for [daylight saving](#)
- Represent time difference between locations using [Coordinated Universal Time \(UTC\)](#)
- Solve practical problems involving UTC

Networks, paths and trees

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies network techniques to solve network problems **MST-11-07**

Related Life Skills outcomes: MA-LS-11, MA-LS-12

Content

Network concepts

- Describe a [network](#) as a collection of objects (vertices) interconnected by lines (edges) that can represent systems in the real world
- Define and use the network terminology: [vertex](#), [edge](#), [degree](#), [directed networks](#) and [weighted edges](#)
- Examine circumstances in which networks are used in a variety of real-world applications
- Construct a network diagram to represent information from a given table or map, showing weighted and directed edges
- Solve problems involving network diagrams in a variety of practical contexts

Shortest paths and spanning trees

- Define and use the network terminology: [path](#), [tree](#), [spanning tree](#) and [minimum spanning tree](#)
- Determine the minimum spanning tree of a network with weighted edges, using any method
- Use minimum spanning trees to solve minimal connector problems
- Identify a [shortest path](#) on a network diagram with no more than 10 vertices
- Describe a circumstance in which a shortest path is not necessarily the best path nor contained in any minimum spanning tree

Data analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- displays and analyses datasets using summary statistics and graphical representations **MST-11-08**

Related Life Skills outcomes: MA-LS-13, MA-LS-14, MA-LS-15

Content

Statistical investigation process

- Identify an issue and pose a question to a targeted [population](#) to gather statistical information
- Develop a survey by applying questionnaire design principles of clear language, unambiguous questions and consideration of number of choices
- Examine issues of privacy, [bias](#), ethics and responsiveness to diverse groups and [cultures](#)

Population and sample

- Compare and contrast systematic sampling, self-selected sampling, random sampling and stratified sampling
- Justify whether a [sample](#) obtained from a population is representative of the population by considering the sampling method
- Describe the potential faults in the design and practicalities of a [data](#) collection process by considering survey design, [experiments](#) and observational studies, and misunderstandings and misrepresentations

Data classification

- Classify and describe [variables](#) as numerical or categorical
- Describe a [numerical variable](#) as [discrete](#) or [continuous](#)
- Describe a [categorical variable](#) as [nominal](#) or [ordinal](#)
- Identify collections of data that can be described as numerical or categorical depending on responses

Display and interpret grouped and ungrouped data

- Recognise and explain why some [datasets](#) need to be grouped to allow for appropriate representation and analysis
- Use a spreadsheet to organise and represent data using appropriate [graphs](#)
- Represent a numerical dataset as either a [frequency distribution](#) table or a [cumulative frequency](#) distribution table and graph the associated histogram with polygon, both with and without using digital tools
- Represent [categorical datasets](#) in tables and column graphs as appropriate, with and without using digital tools
- Select the type of graph best suited to represent various single datasets and justify the choice of graph
- Identify and describe the shape of the distribution of a dataset as either [symmetric](#), positively [skewed](#) or negatively skewed

- Interpret and analyse dot plots, line graphs, sector graphs, [stem-and-leaf plots](#), [back-to-back stem-and-leaf plots](#) and divided bar charts related to real-world applications
- Analyse a statistical [infographic](#) and justify the choice of graphical representations used for the relevant dataset
- Interpret and consider limitations of graphical representations to make conclusions and predictions
- Explain why a given graphical representation can lead to a misinterpretation of data

Measures of centre and spread

- Describe the [mean](#), [median](#) and [mode](#) as [measures of centre](#) and calculate their values for a dataset in graphical form and tabular form, using a scientific calculator and other digital tools
- Identify and describe datasets as uniform, unimodal, bimodal or multimodal
- Identify the [range](#) and [standard deviation](#) as [measures of spread](#) to describe variation in a dataset
- Calculate the range and population standard deviation σ of a dataset using a scientific calculator or other digital tools
- Compare datasets using measures of centre and measures of spread
- Examine the merits of each measure of centre and justify where each measure is most appropriately used
- Identify and describe real-world examples illustrating appropriate and inappropriate uses of measures of centre and measures of spread
- Use a spreadsheet to analyse data including calculating measures of centre and spread

Quartiles and interquartile range

- Determine the [five-number summary](#) from a set of [numerical data](#) or graphical representation
- Determine the [interquartile range \(IQR\)](#) of datasets
- Compare and contrast the use of range and IQR as measures of spread

Five-number summary and box plots

- Represent numerical datasets using a [box plot](#) to display a five-number summary, with and without using digital tools
- Compare and contrast the measures of centre, spread and shape using [parallel](#) box plots
- Determine [quartiles](#) from datasets displayed in histograms and dot plots, and represent these datasets as a box plot
- Interpret box plots to draw conclusions and make inferences about a dataset

Clusters and outliers

- Identify [clusters](#), gaps and [outliers](#) and explain their occurrence in the context of the data
- Apply $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$ to formally identify outliers
- Explain the impact of outliers on the measures of centre and spread

Outcomes and content for Year 12 – Standard 1

Algebraic relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- represents and interprets the relationships between quantities in algebraic and graphical forms **MST-12-S1-01**

Related Life Skills outcomes: MA-LS-01, MA-LS-02, MA-LS-03, MA-LS-04

Content

Simultaneous linear equations

- Graph two [linear equations](#) using a table of values or a graphing application and identify the [point of intersection](#)
- Solve a pair of simultaneous linear equations by finding the point of intersection of their graphs, with and without graphing applications
- Determine and interpret the point of intersection of two [linear](#) graphs in a variety of practical contexts
- Use [simultaneous equations](#) to model and analyse the [break-even point](#) of a problem where [cost](#) and [revenue](#) are represented by linear equations
- Identify the break-even point and solve problems involving [profit](#) and [loss](#) by using a spreadsheet to model the problem

Graphs of practical situations

- Construct a linear or exponential graph using a table of values or graphing application to make predictions in practical contexts
- Represent the description of a real-world situation graphically
- Identify the strengths and limitations of linear and [exponential models](#) in given practical contexts

Investment

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving simple interest and compound interest to make decisions about financial situations **MST-12-S1-02**

Related Life Skills outcomes: MA-LS-05, MA-LS-06

Content

- Calculate [simple interest](#) for different [interest rates](#) and time periods using the formula $I = Prn$ where I = simple interest, P = principal, r = interest rate per time period and n = number of time periods
- Solve problems involving simple interest in a variety of contexts
- Compare simple interest [graphs](#) for different interest rates and time periods, with and without using digital tools
- Calculate the [future value](#), [present value](#) or interest rate of a [compound interest investment](#) using the formula $FV = PV(1 + r)^n$ where FV = future value of the investment, PV = present value of the investment, r = interest rate per time period and n = number of time periods
- Solve problems involving compound interest in a variety of contexts, including [inflation](#) and [appreciation](#)
- Compare the growth of simple interest and compound interest investments either numerically or from a given graph
- Use a spreadsheet to model investments using both simple and compound interest
- Compare and contrast savings accounts, [term deposits](#) and buying property as investment strategies

Depreciation and loans

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving interest and depreciation to make decisions about loans and credit cards **MST-12-S1-03**

Related Life Skills outcomes: MA-LS-05, MA-LS-06

Content

Depreciation

- Apply the [straight-line depreciation](#) method to calculate the [depreciation](#) of an asset using the formula $S = V_0 - Dn$ where S = [salvage value](#), V_0 = initial value of the asset, D = depreciation amount per time period and n = number of time periods
- Apply the declining balance method to calculate the depreciation of an asset using the formula $S = V_0(1 - r)^n$ where S = salvage value, V_0 = initial value of the asset, r = depreciation [rate](#) per time period and n = number of time periods
- Explain the similarities and differences between the declining balance formula and the [compound interest](#) formula
- Compare straight-line depreciation and [declining balance depreciation](#) either numerically or from a given [graph](#)
- Use a spreadsheet to model depreciation using the straight-line and declining balance methods

Loans

- Compare and analyse the [costs](#) associated with [buy now, pay later](#), short term and long term [loans](#)
- Model a [reducing balance loan](#) as an application of compound interest, given the [interest rate](#) per time period, with periodic [repayments](#), using a table for up to three time periods
- Examine the effect of changing the repayment frequency, making additional repayments or a making a [lump-sum payment](#) on the term and cost of repaying a loan
- Use a spreadsheet to model and solve problems involving a reducing balance loan in a variety of contexts

Credit cards

- Explain how [credit cards](#) are an example of a reducing balance loan
- Compare credit card interest rates with interest rates for other loan types
- Identify the various fees and charges associated with credit card usage, and explain the meaning of an [interest-free period](#)
- Interpret a credit card statement, examining the implications of only making the [minimum payment](#)
- Calculate the compound interest charged on a [transaction](#) or outstanding [balance](#) for a given number of days
- Compare credit cards and personal loans as options to meet short-term, medium-term and long-term financial goals

Right-angled triangles

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies Pythagoras' theorem and trigonometry to solve problems involving right-angled triangles **MST-12-S1-04**

Related Life Skills outcomes: MA-LS-07, MA-LS-08, MA-LS-09, MA-LS-10

Content

- Apply [Pythagoras' theorem](#) to solve practical problems in two dimensions
- Apply [trigonometric ratios](#) to find the lengths of unknown sides in right-angled triangles given an [angle](#) in both [degrees](#), and degrees and [minutes](#)
- Apply trigonometric ratios to find the size of unknown angles in right-angled triangles, in both degrees, and degrees and minutes
- Identify and interpret [true bearings](#) and [compass bearings](#) and convert between them
- Solve practical problems involving Pythagoras' theorem, the trigonometry of right-angled triangles, [angles of elevation](#) and [depression](#), and true bearings and compass bearings

Ratios and rates

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- interprets ratios and rates to solve problems in practical contexts **MST-12-S1-05**

Related Life Skills outcomes: MA-LS-07, MA-LS-08, MA-LS-09, MA-LS-10

Content

Ratios

- Recognise [ratios](#) and explain their relationship with fractions
- Express a ratio in simplest form
- Determine the ratio of two quantities
- Divide a quantity in a given ratio
- Solve practical problems involving ratios
- Explain the relationship between ratios and map [scales](#) and apply this relationship to solve problems in a variety of practical contexts
- Obtain measurements from [scale drawings](#) and building plans with commonly used symbols to solve problems in a variety of practical contexts
- Estimate and compare quantities, materials and [costs](#) using measurements taken from scale drawings

Rates

- Explain the difference between ratios and [rates](#)
- Represent given information as a unit rate
- Convert between rates
- Make comparisons between rates in a variety of practical contexts
- Compare [unit prices](#) to determine [best buys](#)
- Use rates to solve problems in a variety of practical contexts
- Interpret and analyse [distance–time graphs](#) to solve problems related to [speed](#), [distance](#) and time
- Examine [fuel consumption rates](#) to compare the efficiency of vehicles

Bivariate data analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- displays and interprets bivariate datasets to solve problems and make predictions **MST-12-S1-06**

Related Life Skills outcomes: MA-LS-13, MA-LS-14, MA-LS-15

Content

Bivariate datasets

- Distinguish between situations involving one-variable [data](#) and [bivariate data](#) and explain when each is needed
- Explain the difference between [variables](#) that show [correlation](#) and those that have a [causal](#) relationship
- Identify the [independent](#) and [dependent variables](#) within a bivariate [dataset](#) where appropriate
- Analyse relationships between independent and dependent variables that may be described as causal

Scatter plots and lines of best fit

- Represent a bivariate dataset using a [scatter plot](#)
- Create a [line of best fit](#) on a scatter plot for a bivariate dataset, by eye and with digital tools
- Describe the form of a dataset as linear or non-linear based on the [association](#) between two variables
- Describe the strength of a [linear relationship](#) between two variables as strong, moderate or weak, and its direction as positive or negative
- Examine lines of best fit to make predictions and recognise limitations of [interpolation](#) and [extrapolation](#) for bivariate datasets within a variety of contexts

Relative frequency and probability

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- models and solves problems involving the probabilities of simple and compound events **MST-12-S1-07**

Related Life Skills outcomes: MA-LS-13, MA-LS-14, MA-LS-15

Content

- Recognise that probabilities range from 0 (impossible) to 1 (certain) and can be represented as fractions, decimals and percentages
- Identify and define the sample space as the set of all possible [outcomes](#) of a [chance experiment](#)
- Determine the number of outcomes for an experiment
- Express the [probability](#) of an [event](#) that has a [finite](#) number of [equally likely outcomes](#), as
$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$
- Apply $P(\text{an event does not occur}) = 1 - P(\text{the event does occur})$ to calculate the probability of the complement of an event
- Represent equally likely outcomes using [arrays](#) and [tree diagrams](#) to solve problems, involving replacement where applicable, in two-stage events
- Use [relative frequency](#) as an estimate of probability
- Recognise that increasing the number of [trials](#) produces relative frequencies that become closer in value to the theoretical probability
- Calculate the [expected frequency](#) of an event occurring using np where n is the number of times an experiment is repeated, and p is the probability that the event occurs each of those times
- Examine the use of statistics and the associated probabilities in shaping decisions made by the media, governments and companies

Outcomes and content for Year 12 – Standard 2

Algebraic relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- represents the relationships between quantities algebraically and graphically to solve problems and make predictions in practical contexts **MST-12-S2-01**

Related Life Skills outcomes: MA-LS-01, MA-LS-02, MA-LS-03, MA-LS-04

Content

Simultaneous linear equations

- Graph two [linear equations](#) and identify the [point of intersection](#), with and without using graphing applications
- Solve a pair of simultaneous linear equations using graphical and algebraic methods
- Develop a pair of simultaneous linear equations to model a practical situation
- Solve practical problems that involve simultaneous linear equations
- Use [simultaneous equations](#) to model and analyse the [break-even point](#) where [cost](#) and [revenue](#) are represented by linear equations
- Identify the break-even point and solve problems involving [profit](#) and [loss](#) using a spreadsheet

Exponential relationships

- Recognise that an [exponential relationship](#) can be represented by an [equation](#), a table of values, a set of ordered pairs or a graph, and move flexibly between these representations
- Graph an exponential relationship of the form $y = a^x$ and $y = a^{-x}$, where $a > 0$, with and without using graphing applications
- Construct and analyse an [exponential model](#) of the form $y = ka^x$ and $y = ka^{-x}$, where $a > 0$ and k is a constant, to solve practical growth or decay problems
- Interpret the meaning of the [intercept](#) of an exponential graph in a variety of contexts
- Explain the limitations of exponential models in practical contexts

Quadratic relationships

- Recognise that a [quadratic relationship](#) can be represented by an equation, a table of values, a set of ordered pairs or a graph, and move flexibly between these representations
- Recognise the parabolic shape of a quadratic relationship, noting its [vertex](#) and [axis of symmetry](#)
- Graph a quadratic relationship of the form $y = ax^2 + bx + c$ using graphing applications
- Identify the x -intercepts and y -intercept of a [parabola](#) from a graph
- Determine the axis of symmetry and vertex of a parabola using the [midpoint](#) of the x -intercepts shown on a graph
- Analyse a given graph of a quadratic relationship and use it to solve practical problems
- Interpret the meaning of the intercepts and vertex of a parabola in a variety of contexts

- Explain the limitations of a [model](#) of a quadratic relationship in a practical context and consider the values for x and y for which the [quadratic model](#) is valid

Reciprocal relationships

- Recognise that [inverse variation](#) is represented by a [reciprocal relationship](#) of the form $y = \frac{k}{x}$, where k is the [constant of variation](#)
- Identify the [hyperbolic](#) shape of a reciprocal graph
- Graph a reciprocal relationship of the form $y = \frac{k}{x}$, with and without using graphing applications
- Construct a reciprocal model of the form $y = \frac{k}{x}$ to analyse inverse variation problems
- Solve inverse variation problems graphically, or algebraically given the value of two of the [variables](#) in $y = \frac{k}{x}$
- Explain the limitations of reciprocal models in practical inverse variation problems

Investment and loans

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- models financial situations and solves problems involving interest, depreciation and borrowing money **MST-12-S2-02**

Related Life Skills outcomes: MA-LS-05, MA-LS-06

Content

Investment

- Calculate [simple interest](#) for different rates and time periods using the formula $I = Prn$ where I = simple interest, P = principal, r = [interest rate](#) per time period and n = number of time periods
- Solve problems involving simple interest in a variety of contexts
- Compare simple interest [graphs](#) for different interest rates and time periods, with and without using digital tools
- Calculate the [future value](#), [present value](#) or interest rate of a [compound interest investment](#) using the formula $FV = PV(1 + r)^n$ where FV = future value of the investment, PV = present value of the investment, r = interest rate per time period and n = number of time periods
- Solve problems involving compound interest in a variety of contexts, including [inflation](#) and [appreciation](#)
- Examine the effect of varying the interest rate, the term or the compounding period on the future value of an investment, with and without using digital tools
- Compare the growth of simple interest and compound interest investments numerically and graphically, with and without using digital tools
- Recognise that simple interest graphs are [linear](#) and compound interest graphs are exponential
- Use a spreadsheet to numerically and graphically model investments using both simple and compound interest
- Interpret and analyse tables and graphs about the value of [share](#)
- Graph and interpret the value of a share over time, with and without using digital tools
- Calculate the [dividend](#) paid and the [dividend yield](#) on shares (excluding franked dividends)
- Solve problems involving calculation of [brokerage costs](#) and total costs of purchasing shares
- Compare and contrast savings accounts, [term deposits](#), shares and buying property as investment strategies

Depreciation

- Apply the [straight-line depreciation](#) method to calculate the [depreciation](#) of an asset using the formula $S = V_0 - Dn$ where S = [salvage value](#), V_0 = initial value of the asset, D = depreciation amount per time period and n = number of time periods
- Apply the declining balance method to calculate the depreciation of an asset using the formula $S = V_0(1 - r)^n$ where S = salvage value, V_0 = initial value of the asset, r = depreciation [rate](#) per time period and n = number of time periods

- Compare straight-line depreciation and [declining balance depreciation](#) both numerically and graphically, with and without using digital tools
- Use a spreadsheet to numerically and graphically model depreciation using the straight-line and declining balance methods

Loans

- Compare and analyse the costs associated with [buy now, pay later](#), short term and long term [loans](#)
- Model a [reducing balance loan](#) as an application of compound interest with periodic [repayments](#) in tabular form for up to four time periods, with and without using digital tools
- Use a spreadsheet to model and solve problems involving a reducing balance loan in a variety of contexts
- Examine the effect of changing the repayment frequency, additional repayments or a [lump-sum payment](#) on the term and cost of a loan using a graph or by calculation
- Solve problems involving reducing balance loans including calculating the total amount paid, monthly repayments, amount still owing and the time taken to repay the loan

Credit cards

- Explain how [credit cards](#) are an example of a reducing balance loan
- Compare credit card interest rates with interest rates for other loan types
- Identify the various fees and charges associated with credit card usage, and understand the meaning of an [interest-free period](#)
- Interpret a credit card statement, examining the implications of only making the [minimum payment](#)
- Calculate the compound interest charged on a purchase, [transaction](#) or outstanding [balance](#) for a given number of days, with and without using digital tools
- Compare credit options to best manage finances to meet short-term, medium-term and long-term goals, such as credit cards and personal loans

Annuities

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving annuities **MST-12-S2-03**

Related Life Skills outcomes: MA-LS-05, MA-LS-06

Content

- Identify an [annuity](#) as an investment account with regular, equal [contributions](#) and [interest](#) compounding at the end of each period, or as a single sum [investment](#) from which regular equal [withdrawals](#) are made
- Model an annuity in tabular form for up to four time periods
- Use a spreadsheet to model an annuity
- Using a table of [interest factors](#), calculate the [future value](#) or [present value](#) of an annuity
- Using a table of interest factors, determine the contribution amount required to achieve a given future value or the single sum contribution required to produce an [equivalent](#) future value for any given annuity
- Examine the effect of varying the amount initially invested, the value of the periodic payment, the [interest rate](#) and the duration of the annuity on the total value of the investment, using digital tools
- Solve annuity problems involving financial decisions regarding [superannuation](#), savings accounts and [loans](#)

Trigonometry

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies trigonometry to solve problems involving right-angled and non-right-angled triangles **MST-12-S2-04**

Related Life Skills outcomes: MA-LS-07, MA-LS-08, MA-LS-09, MA-LS-10

Content

- Apply [trigonometric ratios](#) to right-angled triangles to find either the lengths of unknown sides or the size of unknown [angles](#) in both [degrees](#), and degrees and [minutes](#)
- Apply the [sine rule](#) $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ in a given triangle ABC to find the value of an unknown side
- Apply the sine rule $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$ in a given triangle ABC to find the value of an unknown angle, excluding the [ambiguous case](#)
- Apply the sine rule in a given triangle to find the value of an unknown angle, given that the unknown angle is [obtuse](#)
- Apply the [cosine rule](#) $c^2 = a^2 + b^2 - 2ab \cos C$ in a given triangle ABC to find the value of an unknown side
- Apply the cosine rule $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ in a given triangle ABC to find the value of an unknown angle
- Apply the formula $A = \frac{1}{2}ab \sin C$, where a and b are the sides that form the included angle C in a given triangle ABC , to solve problems in a range of contexts
- Identify and interpret [true bearings](#) and [compass bearings](#) and convert between them
- Construct and interpret compass [radial surveys](#) and solve related problems
- Solve practical problems involving [Pythagoras' theorem](#), the trigonometry of right-angled and non-right-angled triangles, [angles of elevation](#) and [depression](#), and true bearings and compass bearings

Ratios and rates

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves practical problems involving ratios and rates **MST-12-S2-05**

Related Life Skills outcomes: MA-LS-07, MA-LS-08, MA-LS-09, MA-LS-10

Content

Ratios

- Recognise [ratios](#) and explain their relationship with fractions
- Express a ratio in simplest form
- Determine the ratio of two quantities
- Divide a quantity in a given ratio
- Solve practical problems involving ratios
- Solve practical ratio problems involving the [capture-recapture technique](#)
- Explain the relationship between ratios and map [scales](#) and apply this relationship to solve problems in a variety of practical contexts
- Construct [scale drawings](#), with and without using digital tools
- Estimate and compare quantities, materials and [costs](#) using measurements taken from scale drawings and building plans with commonly used symbols
- Calculate the perimeter, or area of a section of land using the [Trapezoidal rule](#) where appropriate, from a site plan, an aerial photograph, a [radial survey](#) or a map that includes a scale
- Calculate the [volume](#) of rainfall over an area using $V = Ah$, where V is the volume, A is the area of land and h is the depth of rain, from a site plan, an aerial photograph, a radial survey or a map that includes a scale

Rates

- Explain the difference between ratios and [rates](#)
- Represent given information as a unit rate
- Convert between rates
- Make comparisons between rates in a variety of practical contexts
- Compare [unit prices](#) to determine [best buys](#)
- Use rates to solve problems in a variety of practical contexts
- Interpret and analyse [distance–time graphs](#) to solve problems related to [speed](#), [distance](#) and time
- Examine [fuel consumption rates](#) to compare the efficiency of vehicles
- Recognise and convert between watts (W) and kilowatts (kW) using $1000\text{ W} = 1\text{ kW}$
- Apply units of energy to calculate [kilowatt hours](#) and solve problems involving the consumption of energy

Network flow

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies network flow concepts to solve problems and model decision-making in practical contexts **MST-12-S2-06**

Related Life Skills outcomes: MA-LS-11, MA-LS-12

Content

- Define and use network flow terminology including [source](#), [sink](#), [outflow](#), [inflow](#) and [cut](#)
- Convert information presented in a table to a weighted and [directed network](#) diagram
- Calculate and analyse the [flow capacity](#) of a [network](#) including the use of the [saturated edges](#) method
- Solve small-scale network flow problems including the use of applying the [maximum-flow minimum-cut theorem](#)
- Examine the impact of increasing or reducing the flow capacity of an [edge](#) in a network diagram
- Explain why the flow capacity of a network is or is not sufficient to meet demand in a variety of contexts

Critical path analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies critical path analysis to solve problems and model decision-making in practical contexts **MST-12-S2-07**

Related Life Skills outcomes: MA-LS-11, MA-LS-12

Content

- Construct a [network](#) diagram from a [precedence table](#) or equivalent, including those involving [dummy activities](#), limited to 10 activities
- Explain the term [critical path](#)
- Apply [forward](#) and [backward scanning](#) to determine the [earliest starting time \(EST\)](#) and [latest starting time \(LST\)](#) for each activity in a project
- Calculate [float times](#) to identify [critical](#) and [non-critical activities](#)
- Analyse ESTs and LSTs to locate the critical path(s) for the project
- Apply critical path analysis to determine the minimum completion time for a project
- Construct a [Gantt chart](#) to represent a project from a network diagram indicating the critical path
- Examine the impact of increasing or reducing the duration of an activity on the completion time of the project or task being modelled
- Use a spreadsheet to model a project involving precedence tables and solve related problems

Bivariate data analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses bivariate datasets using statistical processes **MST-12-S2-08**

Related Life Skills outcomes: MA-LS-13, MA-LS-14, MA-LS-15

Content

Bivariate datasets

- Distinguish between situations involving one-variable [data](#) and [bivariate data](#) and explain when each is needed
- Explain the difference between [variables](#) that show [correlation](#) and those that have a [causal](#) relationship
- Identify the [independent](#) and [dependent variables](#) within a bivariate [dataset](#) where appropriate
- Analyse relationships between independent and dependent variables that may be described as causal

Scatter plots and lines of best fit

- Represent a bivariate dataset using a [scatter plot](#)
- Create a [line of best fit](#) on a scatter plot for a bivariate dataset, by eye and with digital tools
- Describe the form of a dataset as linear or non-linear based on the [association](#) between two variables
- Describe the strength of a [linear relationship](#) between two variables as strong, moderate or weak, and its direction as positive or negative
- Determine and interpret the [intercept](#) and [gradient](#) of the line of best fit from a given [graph](#) to form an [equation](#) of the line
- Calculate and interpret [Pearson's correlation coefficient](#) (r) for a bivariate dataset using a scientific calculator to quantify the strength of a linear association between the two variables
- Determine the equation of the [least-squares regression line](#) for a bivariate dataset using a scientific calculator
- Use a spreadsheet to construct a scatter plot and the least-squares regression line for a bivariate dataset
- Examine lines of best fit to make predictions and recognise limitations of [interpolation](#) and [extrapolation](#) for bivariate datasets within a variety of contexts

Relative frequency and probability

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- models and solves problems involving the probabilities of multistage events **MST-12-S2-09**

Related Life Skills outcomes: MA-LS-13, MA-LS-14, MA-LS-15

Content

- Recognise that probabilities range from 0 (impossible) to 1 (certain) and can be represented as fractions, decimals and percentages
- Identify and define the sample space as the set of all possible [outcomes](#) of a [chance experiment](#)
- Determine the number of outcomes for an experiment
- Express the [probability](#) of an [event](#) that has a [finite](#) number of [equally likely outcomes](#), as
$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$
- Apply $P(\text{an event does not occur}) = 1 - P(\text{the event does occur})$ to calculate the probability of the complement of an event
- Construct and use [arrays](#), [tree diagrams](#) and [probability trees](#) to determine the outcomes and probabilities for [multistage events](#)
- Determine the probabilities of outcomes for multistage events using $P(A \text{ and } B) = P(A) \times P(B)$
- Use [relative frequency](#) as an estimate of probability
- Recognise that increasing the number of [trials](#) produces relative frequencies that become closer in value to the theoretical probability
- Calculate the [expected frequency](#) of an event occurring using np where n is the number of times an experiment is repeated, and p is the probability that the event occurs on each of those times
- Construct and interpret [Venn diagrams](#) and [two-way tables](#) from given information, limited to combinations of two attributes, and use them to solve problems in a variety of contexts
- Examine the use of statistics and the associated probabilities in shaping decisions made by the media, governments and companies

The normal distribution

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses normally distributed datasets using statistical processes **MST-12-S2-10**

Related Life Skills outcomes: MA-LS-13, MA-LS-14, MA-LS-15

Content

Normally distributed datasets

- Recognise that a [dataset](#) that is normally distributed can be represented by a bell-shaped curve
- Explain that the [mean](#), [median](#) and [mode](#) are approximately equal for [data](#) arising from a [random variable](#) that is normally distributed

Calculating z-scores

- Describe the [z -score](#) as the number of [standard deviations](#) that a value is above or below the mean
- Recognise that the set of z -scores for data arising from a random variable that is normally distributed has a mean of 0 and standard deviation of 1
- Calculate the z -score corresponding to a specific value in a dataset by applying the formula $z = \frac{x - \mu}{\sigma}$, where x is a specific value, μ is the mean and σ is the standard deviation
- Use z -scores to compare scores from different datasets and justify conclusions in the context of the problem
- Model and apply the empirical rule, where approximately 68% of data will have z -scores between -1 and 1 , approximately 95% of data will have z -scores between -2 and 2 , and approximately 99.7% of data will have z -scores between -3 and 3

Probability using z-scores

- Calculate probabilities using z -scores and the empirical rule
- Represent probabilities by shading areas under the [normal distribution](#) curve
- Use z -scores to identify probabilities of [events](#) less or more extreme than a given event and solve problems
- Use z -scores to make judgements related to [outcomes](#) of a given event or sets of data